

The Aggregate Opportunity Cost of Recreational Land Use

A Counterfactual Valuation of U.S. Golf Courses

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Abstract

[Placeholder: 228-word summary detailing the macroeconomic inefficiency of golf courses, the HBU methodology, the MICE imputation of 16,297 facilities, the \$244.9B pooled result, and the institutional frictions of zoning and covenants.]

1 Introduction

1.1 Macroeconomic Context and Spatial Misallocation

- **The Problem:** Establish the inelasticity of urban land supply and the resulting "zoning tax" that inflates residential land values.
- **The Inefficiency:** Introduce the preservation of low-density recreational land (golf courses) in these constrained markets as a macroeconomic distortion.

1.2 The Golf Course Typology and Amenity Depreciation

- **Joint Production:** Explain how golf courses historically functioned as surrogate public goods to cross-subsidize residential real estate premiums.
- **Shifting Preferences:** Note the structural decline in golf demand and the shift in consumer preference toward natural open spaces, leaving massive acreage locked in a depreciating use.

1.3 Methodological Framework: Highest and Best Use (HBU)

- **The Counterfactual:** Introduce the appraisal doctrine of HBU and define the core opportunity cost equation: $V_{OC} = \max(V_{Res}, V_{Ag}) - V_{Current}$.
- **The Hypothetical Condition:** Explicitly state the assumption of a frictionless legal environment (dissolving zoning and private covenants) to isolate pure economic value.

1.4 Research Question and Roadmap

- **Core Question:** What is the aggregate financial opportunity cost of U.S. golf courses when evaluated against their HBU counterfactuals?
- **Roadmap:** Briefly outline the progression of the thesis through literature, theory, data (MICE/RUCC), results, and policy implications.

2 Literature Review

- **Urban Land Valuation:** Discuss Davis et al. and Glaeser regarding the "zoning tax" and the extreme spatial concentration of residential land value.
- **Microeconomics of Amenities:** Discuss Rosen, Limehouse, and Cho regarding hedonic pricing, joint production, and the depreciating amenity value of golf courses.
- **Institutional Frictions:** Discuss Coase, Ellickson, and Korngold regarding the failure of market bargaining due to restrictive HOA covenants and exclusionary zoning.

3 Theoretical Framework

- **Neoclassical Land Allocation:** Define spatial equilibrium and the mathematical maximization of residual land value.
- **The Opportunity Cost Function:** Deconstruct the variables V_{Res} , V_{Ag} , and $V_{Current}$.
- **Market Failure:** Model the institutional frictions $F(Z, C)$ (Zoning and Covenants) that prevent V_{OC} from being realized, justifying the Hypothetical Condition.

4 Data and Methodology

- **Dataset Scope:** Detail the 16,297 observations, Tigris spatial boundaries, FHFA residential data, and USDA agricultural data.
- **Hybrid Valuation Algorithm:** Explain the use of USDA RUCC codes to bifurcate the data into Urban (V_{Res}) and Rural (V_{Ag}) counterfactuals to control for elasticity.
- **MICE Imputation:** Detail the Predictive Mean Matching (PMM) protocol used to resolve the 38% missing acreage data across $m = 5$ datasets.
- **Parameter Pooling:** Explain the use of Rubin's Rules to pool the final aggregate sums and regression coefficients to maintain statistical validity.

5 Empirical Results

- **Aggregate Valuation:** Present the pooled \$244.9 billion opportunity cost and its 95% confidence interval.
- **Spatial Heterogeneity:** Present the Urban vs. Rural bifurcation table, highlighting that 98.5% of the value is concentrated in urban areas.
- **Econometric Determinants:** Present the pooled logarithmic regression output, interpreting the coefficients for holes, course type (Public/Private), and urban classification.

6 Discussion and Policy Implications

- **Macroeconomic Impact:** Translate the \$244.9B figure into the real-world cost of foregone housing supply and misallocated labor.
- **The Legal Straitjacket:** Discuss the NIMBY resistance and the "stale covenants" that legally paralyze the land, making Coasian bargaining impossible.
- **Alternative Counterfactuals:** Introduce Weinand's research on utility-scale renewable energy (solar/wind) as a highly synergistic, non-residential HBU.
- **Methodological Limitations:** Acknowledge the "bulk discount" (plattage) limitation of applying subdivided residential lot prices to massive 150-acre parcels.

7 Conclusion

- **Summary:** Restate the \$244.9B finding, the MICE/RUCC methodology, and the extreme urban concentration of the opportunity cost.
- **Final Thought:** Conclude that unlocking this value requires aggressive legal reform to overcome the institutional frictions of zoning and private covenants.